KEMENTERIAN PENDIDIKAN TINGGI, SAINS, DAN TEKNOLOGI
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Mata Kuliah	:	Pemrograman Web Lanjut (PWL)
Program Studi	:	D-IV Teknologi Informasi
Semester	:	4

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Nama	:	Zid'Avwa Al Bari'i
Jobsheet Ke-	:	3

Laporan Jobsheet

Praktikum Ke-1

Langkah	Jawaban/Deskripsi
1	done

Praktikum Ke-2

Langkah	Jawaban/Deskripsi
3	PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan make:migration create_m_level_table --create=m_level INFO Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_02_25_024357_create_m_level_table.php] created successfully. PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS>
4 - 5	PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan migrate INFO Running migrations. 0001_01_01_000000_create_users_table ..... 105.62ms DONE 0001_01_01_000001_create_cache_table ..... 88.15ms DONE 0001_01_01_000002_create_jobs_table ..... 84.26ms DONE 2026_02_25_024357_create_m_level_table ..... 42.51ms DONE
8	PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan make:migration create_m_kategori_table >> php artisan make:migration create_m_supplier_table INFO Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_02_25_025846_create_m_kategori_table.php] created successfully. INFO Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_02_25_025846_create_m_supplier_table.php] created successfully.

database > migrations > 2026_02_25_025846_create_m_kategori_table.php > class > down

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 'm_kategori', callback: function (Blueprint $table): void {
            $table->id(column: 'kategori_id');
            $table->string(column: 'kategori_kode', length: 10)->unique();
            $table->string(column: 'kategori_nama', length: 100);
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 'm_kategori');
    }
};
```

database > migrations > 2026_02_25_025846_create_m_supplier_table.php > class > down

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 'm_supplier', callback: function (Blueprint $table): void {
            $table->id(column: 'supplier_id');
            $table->string(column: 'supplier_kode', length: 10)->unique();
            $table->string(column: 'supplier_nama', length: 100);
            $table->string(column: 'supplier_alamat', length: 255);
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 'm_supplier');
    }
};
```

```
PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan migrate

  INFO  Running migrations.

  2026_02_25_025846_create_m_kategori_table ..................................... 37.66ms DONE
  2026_02_25_025846_create_m_supplier_table ..................................... 35.79ms DONE
```

1 - 3

database > migrations > 2026_03_03_034725_create_m_user_table.php > class

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 'm_user', callback: function (Blueprint $table): void {
            $table->id(column: 'user_id');
            $table->unsignedBigInteger(column: 'level_id')->index;
            $table->string(column: 'username', length: 20)->unique();
            $table->string(column: 'nama', length: 100);
            $table->string(column: 'password');
            $table->timestamps();
        });
    }

```

```
PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan migrate

  INFO  Running migrations.

  2026_03_03_034725_create_m_user_table ..................................................... 176.92ms DONE
```

4

```
PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan make:migration create_m_barang_table
>> php artisan make:migration create_t_penjualan_table
>> php artisan make:migration create_t_stok_table
>> php artisan make:migration create_t_penjualan_detail_table

  INFO  Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_03_03_040330_create_m_barang_table.php] created successfully.

  INFO  Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_03_03_040330_create_t_penjualan_table.php] created successfully.

  INFO  Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_03_03_040331_create_t_stok_table.php] created successfully.

  INFO  Migration [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\migrations\2026_03_03_040332_create_t_penjualan_detail_table.php] created successfully.
```

database > migrations > 2026_03_03_040330_create_m_barang_table.php > class > up > Closure

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 'm_barang', callback: function (Blueprint $table): void {
            $table->id(column: 'barang_id');
            $table->unsignedBigInteger(column: 'kategori_id')->index(); // FK harus index
            $table->string(column: 'barang_kode', length: 10)->unique();
            $table->string(column: 'barang_nama', length: 100);
            $table->integer(column: 'harga_beli');
            $table->integer(column: 'harga_jual');
            $table->timestamps();

            $table->foreign(columns: 'kategori_id')->references(columns: 'kategori_id')->on(table: 'm_kategori');
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 'm_barang');
    }
};
```

database > migrations > 2026_03_03_040330_create_t_penjualan_table.php > class > up > Closure

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 't_penjualan', callback: function (Blueprint $table): void {
            $table->id(column: 'penjualan_id');
            $table->unsignedBigInteger(column: 'user_id')->index();
            $table->string(column: 'pembeli', length: 50);
            $table->string(column: 'penjualan_kode', length: 20)->unique();
            $table->dateTime(column: 'penjualan_tanggal');
            $table->timestamps();

            $table->foreign(columns: 'user_id')->references(columns: 'user_id')->on(table: 'm_user');
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 't_penjualan');
    }
};
```

database > migrations > 2026_03_03_040331_create_t_stok_table.php > ...

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 't_stok', callback: function (Blueprint $table): void {
            $table->id(column: 'stok_id');
            $table->unsignedBigInteger(column: 'supplier_id')->index();
            $table->unsignedBigInteger(column: 'barang_id')->index();
            $table->unsignedBigInteger(column: 'user_id')->index();
            $table->dateTime(column: 'stok_tanggal');
            $table->integer(column: 'stok_jumlah');
            $table->timestamps();

            $table->foreign(columns: 'supplier_id')->references(columns: 'supplier_id')->on(table: 'm_supplier');
            $table->foreign(columns: 'barang_id')->references(columns: 'barang_id')->on(table: 'm_barang');
            $table->foreign(columns: 'user_id')->references(columns: 'user_id')->on(table: 'm_user');
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 't_stok');
    }
};
```

database > migrations > 2026_03_03_040332_create_t_penjualan_detail_table.php > class > up > Closure

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 't_penjualan_detail', callback: function (Blueprint $table): void {
            $table->id(column: 'detail_id');
            $table->unsignedBigInteger(column: 'penjualan_id')->index();
            $table->unsignedBigInteger(column: 'barang_id')->index();
            $table->integer(column: 'harga');
            $table->integer(column: 'jumlah');
            $table->timestamps();

            $table->foreign(columns: 'penjualan_id')->references(columns: 'penjualan_id')->on(table: 't_penjualan');
            $table->foreign(columns: 'barang_id')->references(columns: 'barang_id')->on(table: 'm_barang');
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists(table: 't_penjualan_detail');
    }
};
```

```
2026_03_03_040330_create_m_barang_table ..... 91.67ms DONE
2026_03_03_040330_create_t_penjualan_table ..... 90.62ms DONE
2026_03_03_040331_create_t_stok_table ..... 247.31ms DONE
2026_03_03_040332_create_t_penjualan_detail_table ..... 160.85ms DONE
```

Praktikum Ke-3

Langkah	Jawaban/Deskripsi
1 - 4	

	level_id	level_kode	level_nama	created_at	updated_at
Edit Copy Delete	1	ADM	Administrator	NULL	NULL
Edit Copy Delete	2	MNG	Manager	NULL	NULL
Edit Copy Delete	3	STF	Staff/Kasir	NULL	NULL

5 - 8

	user_id	level_id	username	nama	password
Edit Copy Delete	1	1	admin	Administrator	$2y$12$Vs18C2JBDuX5N0Zydn18SuEZHOBLktu
Edit Copy Delete	2	2	manager	Manager	$2y$12$IGszcnTJFpXDAGQ9oozi TuC5lUDrvC.kH
Edit Copy Delete	3	3	staff	Staff/Kasir	$2y$12$/HFax.hANbfr0ECgxfsqqumHRb7ODYiyr

10

```
PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan make:seeder KategoriSeeder
>> php artisan make:seeder SupplierSeeder
>> php artisan make:seeder BarangSeeder
>> php artisan make:seeder StokSeeder
>> php artisan make:seeder PenjualanSeeder
>> php artisan make:seeder PenjualanDetailSeeder

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\KategoriSeeder.php] created successfully.

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\SupplierSeeder.php] created successfully.

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\BarangSeeder.php] created successfully.

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\StokSeeder.php] created successfully.

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\PenjualanSeeder.php] created successfully.

 INFO  Seeder [D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS\database\seeders\PenjualanDetailSeeder.php] created successfully.
```

database > seeders > KategoriSeeder.php > ...

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

0 references | 0 implementations
class KategoriSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    0 references | 0 overrides
    public function run(): void
    {
        $data = [
            ['kategori_id' => 1, 'kategori_kode' => 'K01', 'kategori_nama' => 'Makanan'],
            ['kategori_id' => 2, 'kategori_kode' => 'K02', 'kategori_nama' => 'Minuman'],
            ['kategori_id' => 3, 'kategori_kode' => 'K03', 'kategori_nama' => 'Elektronik'],
            ['kategori_id' => 4, 'kategori_kode' => 'K04', 'kategori_nama' => 'Pakaian'],
            ['kategori_id' => 5, 'kategori_kode' => 'K05', 'kategori_nama' => 'Peralatan'],
        ];
        DB::table(table: 'm_kategori')->insert(values: $data);
    }
}
```

database > seeders > SupplierSeeder.php > PHP > SupplierSeeder > run()

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

0 references | 0 implementations
class SupplierSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    0 references | 0 overrides
    public function run(): void
    {
        $data = [
            ['supplier_id' => 1, 'supplier_kode' => 'S01', 'supplier_nama' => 'PT Zid', 'supplier_alamat' => 'Malang'],
            ['supplier_id' => 2, 'supplier_kode' => 'S02', 'supplier_nama' => 'CV Avwa', 'supplier_alamat' => 'Surabaya'],
            ['supplier_id' => 3, 'supplier_kode' => 'S03', 'supplier_nama' => 'UD Al Bari', 'supplier_alamat' => 'Jakarta'],
        ];
        DB::table(table: 'm_supplier')->insert(values: $data);
    }
}
```

database > seeders > BarangSeeder.php > PHP Intelephense > BarangSeeder > run

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

0 references | 0 implementations
class BarangSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    0 references | 0 overrides
    public function run(): void
    {
        $data = [];
        for ($i = 1; $i <= 15; $i++) {
            $data[] = [
                'barang_id' => $i,
                'kategori_id' => ($i % 5) + 1, // Kategori ID 1-5 diulang
                'barang_kode' => 'BRG' . str_pad(string: $i, length: 2, pad_string: '0', pad_type: STR_PAD_LEFT),
                'barang_nama' => 'Barang Dummy ' . $i,
                'harga_beli' => 10000,
                'harga_jual' => 15000,
            ];
        }
        DB::table(table: 'm_barang')->insert(values: $data);
    }
}
```

database > seeders > StokSeeder.php > ...

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

0 references | 0 implementations
class StokSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    0 references | 0 overrides
    public function run(): void
    {
        $data = [];
        for ($i = 1; $i <= 15; $i++) {
            $data[] = [
                'stok_id' => $i,
                'supplier_id' => ceil(num: $i / 5), // Barang 1-5 ke Supp 1, 6-10 ke Supp 2, 11-15 ke Supp 3
                'barang_id' => $i,
                'user_id' => 1, // Asumsi diinput oleh Admin
                'stok_tanggal' => now(),
                'stok_jumlah' => 100,
            ];
        }
        DB::table(table: 't_stok')->insert(values: $data);
    }
}
```

database > seeders > PenjualanSeeder.php > ...

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

0 references | 0 implementations
class PenjualanSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    0 references | 0 overrides
    public function run(): void
    {
        $data = [];
        for ($i = 1; $i <= 10; $i++) {
            $data[] = [
                'penjualan_id' => $i,
                'user_id' => 3, // Asumsi diproses oleh Kasir/Staff
                'pembeli' => 'Customer ' . $i,
                'penjualan_kode' => 'PJ' . str_pad(string: $i, length: 2, pad_string: '0', pad_type: STR_PAD_LEFT),
                'penjualan_tanggal' => now(),
            ];
        }
        DB::table(table: 't_penjualan')->insert(values: $data);
    }
}
```

database > seeders > PenjualanDetailSeeder.php > ...

```php
7   use Illuminate\Support\Facades\DB;
8
    0 references | 0 implementations
9   class PenjualanDetailSeeder extends Seeder
10  {
11      /**
12       * Run the database seeds.
13       */
        0 references | 0 overrides
14      public function run(): void
15      {
16          $data = [];
17          $detail_id = 1;
18          for ($i = 1; $i <= 10; $i++) { // 10 transaksi
19              for ($j = 1; $j <= 3; $j++) { // 3 barang per transaksi
20                  $data[] = [
21                      'detail_id' => $detail_id,
22                      'penjualan_id' => $i,
23                      'barang_id' => rand(min: 1, max: 15), // Random barang id 1-15
24                      'harga' => 15000,
25                      'jumlah' => 2,
26                  ];
27                  $detail_id++;
28              }
29          }
30          DB::table(table: 't_penjualan_detail')->insert(values: $data);
31      }
32  }
```

```
PS D:\Kuliah\Pemrogramman\CollegeStudy\WebProgramming\4thSemester\POS> php artisan db:seed

  INFO  Seeding database.

  Database\Seeders\KategoriSeeder ..... RUNNING
  Database\Seeders\KategoriSeeder ..... 28 ms DONE

  Database\Seeders\SupplierSeeder ..... RUNNING
  Database\Seeders\SupplierSeeder ..... 3 ms DONE

  Database\Seeders\BarangSeeder ..... RUNNING
  Database\Seeders\BarangSeeder ..... 4 ms DONE

  Database\Seeders\StokSeeder ..... RUNNING
  Database\Seeders\StokSeeder ..... 3 ms DONE

  Database\Seeders\PenjualanSeeder ..... RUNNING
  Database\Seeders\PenjualanSeeder ..... 4 ms DONE

  Database\Seeders\PenjualanDetailSeeder ..... RUNNING
  Database\Seeders\PenjualanDetailSeeder ..... 4 ms DONE
```

Praktikum Ke-4

Langkah	Jawaban/Deskripsi
1 - 4	

	level_id	level_kode	level_nama	created_at	updated_at
Edit Copy Delete	1	ADM	Administrator	NULL	NULL
Edit Copy Delete	2	MNG	Manager	NULL	NULL
Edit Copy Delete	3	STF	Staff/Kasir	NULL	NULL
Edit Copy Delete	4	CUS	Pelanggan	2026-03-03 05:07:22	NULL

5 - 6	

	level_id	level_kode	level_nama	created_at	updated_at
Edit Copy Delete	1	ADM	Administrator	NULL	NULL
Edit Copy Delete	2	MNG	Manager	NULL	NULL
Edit Copy Delete	3	STF	Staff/Kasir	NULL	NULL
Edit Copy Delete	4	CUS	Customer	2026-03-03 05:07:22	NULL

7	

	level_id	level_kode	level_nama	created_at	updated_at
Edit Copy Delete	1	ADM	Administrator	NULL	NULL
Edit Copy Delete	2	MNG	Manager	NULL	NULL
Edit Copy Delete	3	STF	Staff/Kasir	NULL	NULL

8 - 10	

Data Level Pengguna

ID	Kode Level	Nama Level
1	ADM	Administrator
2	MNG	Manager
3	STF	Staff/Kasir

Praktikum Ke-5

Langkah	Jawaban/Deskripsi
1 - 4	

	kategori_id	kategori_kode	kategori_nama	created_at	updated_at
Edit Copy Delete	1	K01	Makanan	NULL	NULL
Edit Copy Delete	2	K02	Minuman	NULL	NULL
Edit Copy Delete	3	K03	Elektronik	NULL	NULL
Edit Copy Delete	4	K04	Pakaian	NULL	NULL
Edit Copy Delete	5	K05	Peralatan	NULL	NULL
Edit Copy Delete	6	SNK	Snack/Makanan Ringan	2026-03-03 13:47:00	NULL

Click the drop-down arrow to toggle column's visibility.

5 - 6	

	kategori_id	kategori_kode	kategori_nama	created_at	updated_at
Edit Copy Delete	1	K01	Makanan	NULL	NULL
Edit Copy Delete	2	K02	Minuman	NULL	NULL
Edit Copy Delete	3	K03	Elektronik	NULL	NULL
Edit Copy Delete	4	K04	Pakaian	NULL	NULL
Edit Copy Delete	5	K05	Peralatan	NULL	NULL
Edit Copy Delete	6	SNK	Camilan	2026-03-03 13:47:00	NULL

<table>
<tr><td>7</td><td>
<table>
<tr><th></th><th>kategori_id</th><th>kategori_kode</th><th>kategori_nama</th><th>created_at</th><th>updated_at</th></tr>
<tr><td>Edit Copy Delete</td><td>1</td><td>K01</td><td>Makanan</td><td>NULL</td><td>NULL</td></tr>
<tr><td>Edit Copy Delete</td><td>2</td><td>K02</td><td>Minuman</td><td>NULL</td><td>NULL</td></tr>
<tr><td>Edit Copy Delete</td><td>3</td><td>K03</td><td>Elektronik</td><td>NULL</td><td>NULL</td></tr>
<tr><td>Edit Copy Delete</td><td>4</td><td>K04</td><td>Pakaian</td><td>NULL</td><td>NULL</td></tr>
<tr><td>Edit Copy Delete</td><td>5</td><td>K05</td><td>Peralatan</td><td>NULL</td><td>NULL</td></tr>
</table>
</td></tr>
<tr><td>8 - 10</td><td>
<h1>Data Kategori Barang</h1>
<table>
<tr><th>ID</th><th>Kode Kategori</th><th>Nama Kategori</th></tr>
<tr><td>1</td><td>K01</td><td>Makanan</td></tr>
<tr><td>2</td><td>K02</td><td>Minuman</td></tr>
<tr><td>3</td><td>K03</td><td>Elektronik</td></tr>
<tr><td>4</td><td>K04</td><td>Pakaian</td></tr>
<tr><td>5</td><td>K05</td><td>Peralatan</td></tr>
</table>
</td></tr>
</table>

Praktikum Ke-6

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<tr><th>Langkah</th><th>Jawaban/Deskripsi</th></tr>
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<h1>Data User</h1>
<table>
<tr><th>ID</th><th>Username</th><th>Nama</th><th>ID Level Pengguna</th></tr>
<tr><td>1</td><td>admin</td><td>Administrator</td><td>1</td></tr>
<tr><td>2</td><td>manager</td><td>Manager</td><td>2</td></tr>
<tr><td>3</td><td>staff</td><td>Staff/Kasir</td><td>3</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><td>8 - 9</td><td>

Data User

ID	Username	Nama	ID Level Pengguna
1	admin	Administrator	1
2	manager	Manager	2
3	staff	Staff/Kasir	3
4	customer-1	Pelanggan	4

</td></tr>
<tr><td>10 - 11</td><td>

Data User

ID	Username	Nama	ID Level Pengguna
1	admin	Administrator	1
2	manager	Manager	2
3	staff	Staff/Kasir	3
4	customer-1	Pelanggan Pertama	4

</td></tr>
</table>

Soal Praktikum

Langkah	Jawaban/Deskripsi
1	used to secure and encrypt data within your Laravel application, such as user sessions and cookies.
2	generate it by running the php artisan key:generate command in terminal.
3	Laravel provides 4 migration files. They are used to create the default tables needed for user authentication and background system processes (users, password resets, failed jobs, and access tokens).
4	It automatically creates created_at and updated_at columns in your table to track exactly when a row is inserted or modified.
5	It generates a bigint(20) unsigned data type that automatically increments.

6	Using $table->id(); creates a primary key column named exactly id. Adding a string inside, like $table->id('level_id');, creates the primary key but customizes the column name to level_id.
7	ensures that every value entered into that column is different, preventing duplicate data.
8	In the m_level table, $table->id('level_id') is used to create the actual auto-incrementing Primary Key. In the m_user table, $table->unsignedBigInteger('level_id') is used to create a regular column that acts as a Foreign Key. The Foreign Key must strictly match the data type of the Primary Key it points to.
9	The Hash class is used to securely encrypt (hash) passwords. The code Hash::make('1234') takes the plain text '1234' and turns it into a secure, scrambled hash before saving it to the database.
10	It acts as a placeholder (parameter binding). It safely passes your values into the SQL statement, which protects your database from SQL injection attacks.
11	tells the Model exactly which database table it belongs to. protected $primaryKey = 'user_id'; defines the custom primary key of that table since it's not using the default id.
12	Eloquent ORM is generally considered the easiest. It treats each table row as a PHP object